Project Research Notebook



ACE

CYBER R&D Program

Denzel Jefferson

PI: Charleston Southern University

University: Contextualized Threat Intelligence Prioritization for CTI Analysts

Project Title:

Execution ID: RUN-20260502-001

Date: 2026-05-02

Start Time (UTC): 11:00:00 AM

End Time (UTC): 05:00:00 PM

Researcher(s): Denzel Jefferson

A. Linked Documents:

Test Plan: TP-001-v1.0

Test Case(s): TC-001 *Baseline Log processing without automated threat prioritization*

B. Objective:

- *To measure system performance and analyst workload without automated threat prioritization.*

C. Summary of Actions:

- **Started Ubuntu testbed environment**
- **Verified Elasticsearch, Kibana, and Filebeat services**
- **Generated a synthetic dataset (1000 Logs) from the host VM**
- **Ingested logs into Filebeat**
- **Manually reviewed logs using Kibana Discover**
- **Measured time required to identify high-priority threats**

[High-level narrative of the steps taken. e.g., "Reverted all VMs to 'CLEAN_BASELINE' snapshot. Verified services were running. Started data collectors. Executed attack script from `attacker-vm` per TC-001. Monitored target resource usage. Stopped collectors and attack script after 300s. Powered down testbed."]

D. Key Observations (Qualitative Data):

- **Logs appeared unfiltered**
- **High alert noise**
- **No alert prioritization available**
- **Manual scrolling required**
- **An improper time window initially caused missing logs**

[NOTE: This is for human observations that data alone won't show.]

[e.g., "Noticed an unusual amount of ARP traffic from the target that was not expected."]

E. Anomalies, Errors, and Deviations from Test Plan:

- **The time window was initially selected incorrectly**
- **Adjusted time window from 15 minutes to 15 hours**

[Did anything go wrong or differ from the plan? This is CRITICAL.]

[e.g., "Deviation: Forgot to start tcpdump on `router-vm` until 30 seconds into the test. This time delta must be accounted for during analysis."]

F. Testbed State Changes:

- **None**

[Did you have to make a change to the testbed config during the test? All changes MUST be logged in the formal Change Management Log.]

[e.g., "None."]

G. Data Pointers:

Total Logs Observed: 1000

- **Failed Log Events: 500**
- **Multiple Failed Log Events: 300**
- **Normal Log Events: 200**

Raw Data Location: *[Path to the directory containing all data for this run, e.g., `/data/raw/RUN-20231027-001/`]*

Data Integrity Checksums (SHA256):

Example Entries…

“victim-vm_syslog.log`: [checksum]

H. Next Steps / Follow-up Actions:

[e.g., "Analysis of `victim-vm_syslog.log` required to diagnose the cause of the webserver crash."]

I. Accuracy Metrics

Priority	Expected Logs	Detected Logs	Accuracy

High	300	300	100%
Medium	500	500	100%
Low	200	200	100%

J. Measurement Methodology

Threat identification time is measured using stopwatch timing from log ingestion to the identification of high-priority threats.

K. Experiment Runs

Experiment Runs	Time (sec)
Experiment Run 1	3120 (52 mins)
Experiment Run 2	3060 (51 mins)
Experiment Run 3	2880 (48 mins)
Experiment Run 4	3420 (57 mins)
Experiment Run 5	3240 (54 mins)

Average Time: 3144 seconds (~52.4 minutes)

Execution ID: RUN-20260502-002

Date: 2026-05-02

Start Time (UTC): 05:30:00 PM

End Time (UTC): 05:45:00 PM

Researcher(s): Denzel Jefferson

A. Linked Documents:

Test Plan: TP-001-v1.2

Test Case(s): TC-002 *Automated threat intelligence prioritization*

B. Objective:

- *To evaluate the effectiveness of using automated prioritization with a weighted contextual scoring model that simulates MISP-style scoring.*

[A brief, one-sentence goal for this specific run. e.g., "Measure baseline performance under no-load conditions," or "Execute a 10k pps SYN flood against the webserver."]

C. Summary of Actions:

- **Started Ubuntu testbed environment**
- **Verified Elasticsearch, Kibana, and Filebeat services**
- **Created a runtime field, Automated Priority, in Kibana Data Views**
- **Implemented a weighted scoring-based prioritization logic using severity indicators and contextual metadata.**
- **Executed log generation script ./generate_logs.sh**
- **Logs ingested via Filebeat into Elasticsearch**
- **Applied automated filtering using Kibana Discover**
- **Measured time required to identify high-priority threats**

[High-level narrative of the steps taken. e.g., "Reverted all VMs to 'CLEAN_BASELINE' snapshot. Verified services were running. Started data collectors. Executed attack script from `attacker-vm` per TC-001. Monitored target resource usage. Stopped collectors and attack script after 300s. Powered down testbed."]

D. Key Observations (Qualitative Data):

- **Logs were automatically placed into priority labels**
- **High-priority threats were immediately visible**
- **Analyst workload was greatly reduced**
- **Reduced alert noise compared to baseline manual review**
- **Automated filtering improved efficiency**
- **Results were consistent across multiple runs with low signs of inconsistency**

[NOTE: This is for human observations that data alone won't show.]

[e.g., "Noticed an unusual amount of ARP traffic from the target that was not expected."]

E. Anomalies, Errors, and Deviations from Test Plan:

- **Initial runtime field produced “UNKNOWN” values**
- **Script was adjusted to normalize case (lowercase & uppercase handling)**

[Did anything go wrong or differ from the plan? This is CRITICAL.]

[e.g., "Deviation: Forgot to start tcpdump on `router-vm` until 30 seconds into the test. This time delta must be accounted for during analysis."]

F. Testbed State Changes:

- **Created a runtime field, Automated priority, in Kibana Data Views**

[Did you have to make a change to the testbed config during the test? All changes MUST be logged in the formal Change Management Log.]

[e.g., "None."]

G. Data Pointers:

Total Logs Observed: 1065

- **High Priority Events: 28.2%**
- **Medium Priority Events: 28.2%**
- **Low Priority Events: 43.7%**

Raw Data Location: *[Path to the directory containing all data for this run, e.g., `/data/raw/RUN-20231027-001/`]*

Data Integrity Checksums (SHA256):

“victim-vm_syslog.log`: [checksum]

H. Next Steps / Follow-up Actions:

[e.g., "Analysis of `victim-vm_syslog.log` required to diagnose the cause of the webserver crash."]

I. Accuracy Metrics

Priority	Expected Logs	Detected Logs	Accuracy
High	200	200	100%
Medium	300	300	100%
Low	500	500	100%

J. Measurement Methodology

- Threat identification time is measured from log ingestion to identification using automated filtering in Kibana Discover

K. Experiment Runs

Experiment Runs	Time
Experiment Run 1	10 seconds
Experiment Run 2	6 seconds
Experiment Run 3	5 seconds
Experiment Run 4	6 seconds
Experiment Run 5	7 seconds

- Average Time: 6.8 seconds

Execution ID: RUN-20260502-003

Date: 2026-05-02

Start Time (UTC): 06:00:00 PM

End Time (UTC): 06:10:00 PM

Researcher(s): Denzel Jefferson

A. Linked Documents:

Test Plan: TP-001-v1.2

Test Case(s): TC-003 High-volume Log test

B. Objective:

- *Evaluates the performance and scalability of automated prioritization under increased log volume.*

[A brief, one-sentence goal for this specific run. e.g., "Measure baseline performance under no-load conditions," or "Execute a 10k pps SYN flood against the webserver."]

C. Summary of Actions:

- **Started Ubuntu testbed environment**
- **Verified Elasticsearch, Kibana, and Filebeat services**
- **Executed the log generation script multiple times**
- **Generated a synthetic dataset (2000 Logs) from the host VM**
- **Logs ingested via Filebeat into Elasticsearch**
- **Applied automated filtering using Kibana Discover**
- **Measured identification time under increased volume load**

[High-level narrative of the steps taken. e.g., "Reverted all VMs to 'CLEAN_BASELINE' snapshot. Verified services were running. Started data collectors. Executed attack script

from `attacker-vm` per TC-001. Monitored target resource usage. Stopped collectors and attack script after 300s. Powered down testbed."]

D. Key Observations (Qualitative Data):

- **Automated prioritization remained effective**
- **Increased log volume slightly increased processing time**
- **High-priority threats remained easily identifiable**
- **The system remained functional during high-volume testing**

[NOTE: This is for human observations that data alone won't show.]

[e.g., "Noticed an unusual amount of ARP traffic from the target that was not expected."]

E. Anomalies, Errors, and Deviations from Test Plan:

- **Slight delay observed during log ingestion**

[Did anything go wrong or differ from the plan? This is CRITICAL.]

[e.g., "Deviation: Forgot to start tcpdump on `router-vm` until 30 seconds into the test. This time delta must be accounted for during analysis."]

F. Testbed State Changes:

- **None**

[Did you have to make a change to the testbed config during the test? All changes MUST be logged in the formal Change Management Log.]

[e.g., "None."]

G. Data Pointers:

Total Logs Observed: 2000

- **High Priority Events: 400**
- **Medium Priority Events: 600**
- **Low Priority Events: 1000**

Raw Data Location: *[Path to the directory containing all data for this run, e.g., `/data/raw/RUN-20231027-001/`]*

Data Integrity Checksums (SHA256):

“victim-vm_syslog.log`: [checksum]

H. Next Steps / Follow-up Actions:

[e.g., "Analysis of `victim-vm_syslog.log` required to diagnose the cause of the webserver crash."]

I. Accuracy Metrics

Priority	Expected Logs	Detected Logs	Accuracy
High	400	400	100%
Medium	600	600	100%
Low	1000	1000	100%

J. Measurement Methodology

- Threat identification time is measured from log ingestion to threat identification using automated filtering in Kibana Discover

K. Experiment Runs

Experiment Runs	Time
Experiment Run 1	7 seconds
Experiment Run 2	4 seconds
Experiment Run 3	6 seconds
Experiment Run 4	8 seconds
Experiment Run 5	4 seconds

- Average Time: 5.8 seconds